Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $\cos\alpha+\cos\beta=a, \sin\alpha+\sin\beta=b$ and $\alpha-\beta=2\theta$, then $\frac{\cos 3\theta}{\cos\theta}=$

(A) a^2+b^2-2

(B) a^2+b^2-3

(C) $3-a^2-b^2$

(D) $\left(a^2+b^2\right)/4$

Q.2

$\sin^2\left(\frac{\pi}{8}+\frac{A}{2}\right)-\sin^2\left(\frac{\pi}{8}-\frac{A}{2}\right)$ is equal to:

(A) $(1/\sqrt{2})\sin A$

(B) $(1/\sqrt{2})\cos A$

(C) $\sqrt{2}\sin A$

(D) $\sqrt{2}\cos A$

Q.3

$15[\tan 2\theta+\sin 2\theta]+8=0$ if

(A) $\tan\theta=1/2$

(B) $\sin\theta=1/4$

(C) $\tan\theta=2$

(D) $\cos\theta=1/5$

Q.4

If A and B be acute positive angles satisfying $3\sin^2 A+2\sin^2 B=1, 3\sin 2A-2\sin 2B=0$ then

(A) $B=\pi/4-A/2$

(B) $A=\pi/4-2B$

(C) $B=\pi/2-A/4$

(D) $A=\pi/4-B/2$

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Q.5

If $\theta = \pi/8$, and $f(\theta) = \cos^6\theta + \cos^6 3\theta + \cos^6 5\theta + \cos^6 7\theta$ then $f(\theta)$ is equal to:

(A) $3/4$

(B) 1

(C) $5/4$

(D) $3/2$

Q.6

If an angle θ is divided into 2 parts A and B such that $\mathrm{A} - \mathrm{B} = k$ and $\mathrm{A} + \mathrm{B} = \theta$ and $\tan\mathrm{A} : \tan\mathrm{B} = k : 1$, then the
value of $\sin k$ is :

(A) $\frac{k+1}{k-1}\sin\theta$

(B) $\frac{k}{k+1}\sin\theta$

(C) $\frac{k-1}{k+1}\sin\theta$

(D) None of these

Q.7

If for $n \in \mathrm{N}$,
$f_n(\theta) = \tan(\theta/2)(1+\sec\theta)(1+\sec 2\theta)(1+\sec 4\theta)\ldots$
$(1+\sec 2^n\theta)$, then correct statement is

(A) $f_2(\pi/16) = 1$

(B) $f_3(\pi/32) = 1$

(C) $f_4(\pi/64) = 1$

(D) All of these

Q.8

If A and B are positive acute angles satisfying
$3\cos^2\ \mathrm{A} + 2\cos^2\ \mathrm{B} = 4$ and $\frac{3\sin\mathrm{A}}{\sin\mathrm{B}} = \frac{2\cos\mathrm{B}}{\cos\mathrm{A}}$,
Then the value of $\mathrm{A} + 2\ \mathrm{B}$ is equal to :

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{2}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{4}$

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Q.9

If $\frac{\sin(x+y)}{\sin(x-y)} = \frac{a+b}{a-b}$, then $\frac{\tan x}{\tan y}$ is equal to

(A) $\frac{b}{a}$

(B) $\frac{a}{b}$

(C) ab

(D) None of these

Q.10

If $\sin\theta = 3\sin(\theta + 2\alpha)$, then the value of $\tan(\theta + \alpha) + 2\tan\alpha$ is

(A) 3

(B) 2

(C) -1

(D) 0

Q.11

The value of $\tan A + \tan(60^\circ + A) - \tan(60^\circ - A)$ is

(A) $\tan 3\ \mathrm{A}$

(B) $2\tan 3\ \mathrm{A}$

(C) $3\tan 3\ \mathrm{A}$

(D) None of these

Q.12

What is the value of $\left(1+\cos\frac{\pi}{8}\right)\left(1+\cos\frac{3\pi}{8}\right)\left(1+\cos\frac{5\pi}{8}\right)\left(1+\cos\frac{7\pi}{8}\right)$?

(A) $\frac{1}{2}$

(B) $\frac{1}{2} + \frac{1}{2\sqrt{2}}$

(C) $\frac{1}{2} - \frac{1}{2\sqrt{2}}$

(D) $\frac{1}{8}$

Q.13

What is $\frac{\cot 224^\circ - \cot 134^\circ}{\cot 226^\circ + \cot 316^\circ}$ equal to ?

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(A) $-\operatorname{cosec} 88^{\circ}$

(B) $-\operatorname{cosec} 2^{\circ}$

(C) $-\operatorname{cosec} 44^{\circ}$

(D) $-\operatorname{cosec} 46$

Q.14

If $\sin 18^{0}=\frac{\sqrt{5}-1}{4}$, then what is the value of $\sin 81^{\circ}$?

(A) $\frac{\sqrt{3+\sqrt{5}}+\sqrt{5-\sqrt{5}}}{4}$

(B) $\frac{\sqrt{3+\sqrt{5}}+\sqrt{5+\sqrt{5}}}{4}$

(C) $\frac{\sqrt{3-\sqrt{5}}+\sqrt{5-\sqrt{5}}}{4}$

(D) $\frac{\sqrt{3+\sqrt{5}}-\sqrt{5-\sqrt{5}}}{4}$

Q.15

If $3 \sin \theta+5 \cos \theta=5$, then the value of $5 \sin \theta-3 \cos \theta$ is equal to

(A) 5

(B) 3

(C) 4

(D) none of these

Q.16

Three expressions are given below :

$Q_{1}=\sin (A+B)+\sin (B+C)+\sin (C+A)$

$Q_{2}=\cos (A-B)+\cos (B-C)+\cos (C-A)$

$Q_{3}=\sin A(\cos B+\cos C)+\sin B(\cos C+\cos A)+$
$\sin C(\cos A+\cos B)$

Which one of the following is correct?

(A) $Q_{1}=Q_{2}$

(B) $Q_{2}=Q_{3}$

(C) $Q_{1}=Q_{3}$

(D) All the expressions are different

Q.17

If $\mathrm{A}=(\cos 12^{\circ}-\cos 36^{\circ})(\sin 96^{\circ}+\sin 24^{\circ})$ and

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$\mathrm{B} = (\sin 60^\circ - \sin 12^\circ)(\cos 48^\circ - \cos 72^\circ)$, then what is $\frac{\mathrm{A}}{\mathrm{B}}$ equal to?

(A) -1

(B) 0

(C) 1

(D) 2

Q.18

If $\alpha, \beta, \gamma \in \left(0, \frac{\pi}{2}\right)$, then $\frac{\sin(\alpha+\beta+\gamma)}{\sin\alpha+\sin\beta+\sin\gamma}$ is

(A) < 1

(B) > 1

(C) $= 1$

(D) None of these

Q.19

If $0 < x < \pi$ and $\cos x + \sin x = \frac{1}{2}$, then $\tan x$ is

(A) $\frac{(1-\sqrt{7})}{4}$

(B) $\frac{(4-\sqrt{7})}{3}$

(C) $-\frac{(4+\sqrt{7})}{3}$

(D) $\frac{(1+\sqrt{7})}{4}$

Q.20

$\sin A + 2\sin 2A + \sin 3A$ is equal to which of the following?

1. $4\sin 2A\cos^2\left(\frac{A}{2}\right)$

$2\sin 2\ \mathrm{A}\left(\sin\frac{\mathrm{A}}{2} + \cos\frac{\mathrm{A}}{2}\right)^2$

3. $8\sin A\cos A\cos^2\left(\frac{A}{2}\right)$

Select the correct answer using the code given below:

(A) 1 and 2 only

(B) 2 and 3 only

(C) 1 and 3 only

(D) 1,2 and 3

Answers & Solutions

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Q.1 Answer:

$a^2 + b^2 - 3$

Solution:

From the given relations we have

$$\begin{aligned} a^2 + b^2 &= \cos^2\alpha + \cos^2\beta + 2\cos\alpha\cos\beta + \sin^2\alpha + \\ &\quad \sin^2\beta + 2\sin\alpha\sin\beta \\ &= 2 + 2\cos(\alpha - \beta) \\ &= 2 + 2\cos 2\theta = 4\cos^2\theta \end{aligned}$$

$$\text{Now } \frac{\cos 3\theta}{\cos\theta} = \frac{4\cos^3\theta - 3\cos\theta}{\cos\theta}$$

$$= 4\cos^2\theta - 3 = a^2 + b^2 - 3.$$

Q.2 Answer:

$(1/\sqrt{2})\sin A$

Solution:

Use : $\sin^2 A - \sin^2 B = \sin(A - B)\sin(A + B)$

Q.3 Answer:

$\tan\theta = 2$

Solution:

We have $\tan 2\theta + \sin 2\theta = -8/15$

$$\Rightarrow \quad \frac{2\tan\theta}{1-\tan^2\theta} + \frac{2\tan\theta}{1+\tan^2\theta} = \frac{-8}{15}$$

$$\Rightarrow \quad 15 \times 4\tan\theta + 8\left(1 - \tan^4\theta\right) = 0$$

$$\Rightarrow \quad 8\tan^4\theta + 60\tan\theta - 8 = 0$$

which is satisfied if $\tan\theta = 2$

Q.4 Answer:

$B = \pi/4 - A/2$

Solution:

$\sin 2B = (3/2)\sin 2A, \cos 2B = 3\sin^2 A.$

$\Rightarrow \tan 2B = \cot A \Rightarrow \tan A \tan 2B = 1$

$\Rightarrow A + 2B = \pi/2$

$\Rightarrow B = \pi/4 - A/2$

Q.5 Answer:

$5/4$

Solution:

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$3\theta = \pi/2 - \pi/8, 5\theta = \pi/2 + \pi/8, 7\theta = \pi - \pi/8$, **we get**

$$
\begin{aligned}
f(\theta) =& 2\left(\cos^6\theta + \sin^6\theta\right) \\
=& 2\left[\left(\cos^2\theta + \sin^2\theta\right)^3 - 3\cos^2\theta\sin^2\theta\left(\cos^2\theta + \sin^2\theta\right)\right] \\
=& 2\left[1 - \frac{3}{4}(2\cos\theta\sin\theta)^2\right] = 2\left[1 - \frac{3}{4}\left(\frac{1}{\sqrt{2}}\right)^2\right] \\
=& 5/4
\end{aligned}
$$

Q.6 Answer:

$\frac{k-1}{k+1}\sin\theta$

Solution:

Given an angle θ which is divided into two parts A and B such that $A - B = k$ and $A + B = \theta$ and $\tan A : \tan B = k : 1$, i.e. $\frac{\tan A}{\tan B} = \frac{k}{1}$

$\Rightarrow \frac{\tan A + \tan B}{\tan A - \tan B} = \frac{k+1}{k-1}$

(by componendo and dividendo) $\Rightarrow \frac{\sin(A+B)}{\sin(A-B)} = \frac{k+1}{k-1} \quad \Rightarrow \frac{\sin\theta}{\sin k} = \frac{k+1}{k-1}$

$\Rightarrow \sin k = \frac{k-1}{k+1}\sin\theta$

Q.7 Answer:

All of these

Solution:

$$
\begin{aligned}
\text{Consider } \tan\frac{\theta}{2}(1+\sec\theta) &= \tan\frac{\theta}{2}\left(\frac{1+\cos\theta}{\cos\theta}\right) \\
&= \frac{\sin\theta/2}{\cos\theta/2} \cdot \frac{2\cos^2\theta/2}{\cos\theta} \ldots = \frac{\sin\theta}{\cos\theta} = \tan\theta \\
\therefore f_1(\theta) &= \tan\theta/2(1+\sec\theta)(1+\sec 2\theta) \\
&= [\tan\theta/2(1+\sec\theta)](1+\sec 2\theta) \\
&= (\tan\theta)(1+\sec 2\theta) \quad [\text{ from (1)}] \\
&= \tan 2\theta \quad [\text{ replacing } \theta \text{ by } 2\theta \text{ as above}]
\end{aligned}
$$

$\Rightarrow f_1(\theta) = \tan 2^1\theta$

Similarly, $f_2(\theta) = \tan 2^2\theta, f_3(\theta)$

$= \tan 2^3\theta, f_4(\theta) = \tan 2^4\theta$ **etc.**

$\Rightarrow \quad f_2\left(\frac{\pi}{16}\right) = \tan\left(2^2\frac{\pi}{16}\right) = \tan\frac{\pi}{4} = 1$

$f_3\left(\frac{\pi}{32}\right) = \tan\left(2^3\frac{\pi}{32}\right) = \tan\frac{\pi}{4} = 1$

$f_4\left(\frac{\pi}{64}\right) = \tan\left(2^4\frac{\pi}{64}\right) = \tan\frac{\pi}{4} = 1$

Q.8 Answer:

$\frac{\pi}{2}$

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Solution:

Given, $3\cos^2 \mathrm{A} + 2\cos^2 \mathrm{B} = 4$

$\Rightarrow 2\cos^2 B - 1 = 4 - 3\cos^2 A - 1$

$\Rightarrow \cos 2B = 3\left(1 - \cos^2 A\right) = 3\sin^2 A$

and $2\cos \mathrm{B}\sin \mathrm{B} = 3\sin \mathrm{A}\cos \mathrm{A}$

$\sin 2B = 3\sin A\cos A$

Now, $\cos(\mathrm{A} + 2\,\mathrm{B}) = \cos \mathrm{A}\cos 2\,\mathrm{B} - \sin \mathrm{A}\sin 2\,\mathrm{B}$

$= \cos A\left(3\sin^2 A\right) - \sin A(3\sin A\cos A) = 0$

[using eqs. (1) and

$\Rightarrow \mathrm{A} + 2\,\mathrm{B} = \frac{\pi}{2}$

Q.9 Answer:

$\frac{a}{b}$

Solution:

Let $\frac{\sin(x+y)}{\sin(x-y)} = \frac{a+b}{a-b}$

If $\frac{a}{b} = \frac{c}{d}$, then by componendo and dividendo, we have $\frac{a+b}{a-b} = \frac{c+d}{c-d}$.

Applying componendo and dividendo, we get $\frac{\sin(x+y)+\sin(x-y)}{\sin(x+y)-\sin(x-y)} = \frac{(a+b)+(a-b)}{(a+b)-(a-b)}$

$\Rightarrow \quad \frac{2\sin x\cos y}{2\cos x\sin y} = \frac{2a}{2b}$

[using $\sin(A+B)$ and $\sin(A-B)$]

$\Rightarrow \quad \frac{\tan x}{\tan y} = \frac{a}{b}$.

Q.10 Answer:

0

Solution:

$\sin\theta = 3\sin(\theta + 2\alpha) \Rightarrow \sin(\theta + \alpha - \alpha) = 3\sin$

$(\theta + \alpha + \alpha)$

$\Rightarrow \sin(\theta+\alpha)\cos\alpha - \cos(\theta+\alpha)\sin\alpha$

$= 3\sin(\theta+\alpha)\cos\alpha + 3\cos(\theta+\alpha)\sin\alpha$

$\Rightarrow -2\sin(\theta+\alpha)\cos\alpha = 4\cos(\theta+\alpha)\sin\alpha$

$\Rightarrow \frac{-\sin(\theta+\alpha)}{\cos(\theta+\alpha)} = \frac{2\sin\alpha}{\cos\alpha} \Rightarrow \tan(\theta+\alpha) + 2\tan\alpha = 0$

Q.11 Answer:

$3\tan 3\ \mathrm{A}$

Solution:

The given expression

$= \tan \mathrm{A} + \left[\frac{\sqrt{3}+\tan \mathrm{A}}{1-\sqrt{3}\tan \mathrm{A}}\right] - \left[\frac{\sqrt{3}-\tan \mathrm{A}}{1+\sqrt{3}\tan \mathrm{A}}\right]$

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$= \tan A + \left[\frac{8\tan A}{1-3\tan^2 A}\right] = \frac{9\tan A - 3\tan^3 3}{1-3\tan^2 A}$

$= 3 \cdot \frac{\left(3\tan A - \tan^3 A\right)}{1-3\tan^2 A} = 3\tan 3A$

Q.12 Answer:

$\frac{1}{8}$

Solution:

$\left[1+\cos\frac{\pi}{8}\right]\left[1+\cos\frac{3\pi}{8}\right]\left[1+\cos\frac{5\pi}{8}\right]\left[1+\frac{\cos 7\pi}{8}\right]$

We have, $\cos\frac{7\pi}{8} = \cos\left[\pi - \frac{\pi}{8}\right] = -\cos\frac{\pi}{8}$

and $\cos\frac{5\pi}{8} = \cos\left[\pi - \frac{3\pi}{8}\right] = -\cos\frac{3\pi}{8}$

$\therefore = \left[1+\cos\frac{\pi}{8}\right]\left[1+\cos\frac{3\pi}{8}\right]\left[1-\cos\frac{\pi}{8}\right]\left[1-\cos\frac{3\pi}{8}\right]$

$= \left[1-\cos^2\frac{\pi}{8}\right]\left[1-\cos^2\frac{3\pi}{8}\right] = \sin^2\frac{\pi}{8}\cdot\sin^2\frac{3\pi}{8}$

$= \frac{1}{4}\left[2\sin^2\frac{\pi}{8}\cdot 2\sin^2\frac{3\pi}{8}\right]$

$= \frac{1}{4}\left[\left(1-\cos\frac{\pi}{4}\right)\left(1-\cos\frac{3\pi}{4}\right)\right]$

$\left(\because 1-\cos\theta = 2\sin^2\frac{\theta}{2}\right)$

$= \frac{1}{4}\left[\left(1-\frac{1}{\sqrt{2}}\right)\left(1+\frac{1}{\sqrt{2}}\right)\right] = \frac{1}{8}$

Q.13 Answer:

$-\operatorname{cosec} 2^\circ$

Solution:

$\frac{\cot 224^\circ - \cot 134^\circ}{\cot 226^\circ + \cot 316^\circ}$

$= \frac{\cot(180^\circ+44^\circ) - \cot(180^\circ-46^\circ)}{\cot(180+46^\circ)+\cot(270^\circ+46)^\circ}$

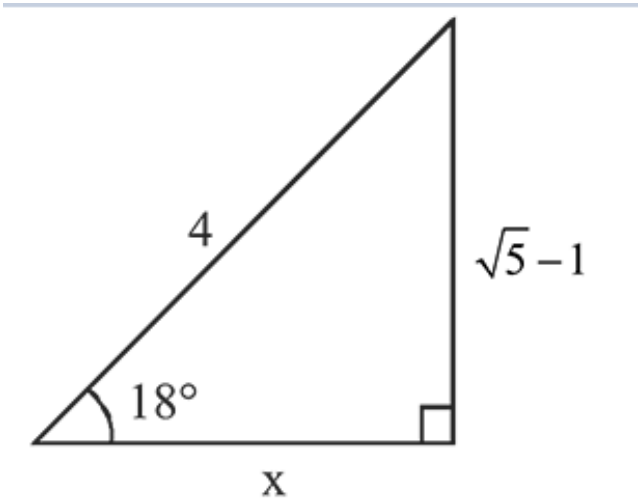
$= \frac{\cot 44^\circ + \cot 46^\circ}{\cot 46^\circ - \tan 46^\circ} = \frac{\tan 46^\circ + \tan 44^\circ}{\tan 44^\circ - \tan 46^\circ}$

$= \frac{\sin(46^\circ+44^\circ)}{\sin(44^\circ-46^\circ)} = -\operatorname{cosec} 2^\circ$

Q.14 Answer:

$\frac{\sqrt{3+\sqrt{5}}+\sqrt{5-\sqrt{5}}}{4}$

Solution:



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$\sin 18^\circ = \frac{\sqrt{5}-1}{4}$

$x^2 = 4^2 - (\sqrt{5}-1)^2$

$\Rightarrow \quad x = \sqrt{10+2\sqrt{5}}$

$\Rightarrow \cos 18^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4}$

$\Rightarrow \quad 2\cos^2 9 - 1 = \frac{\sqrt{10+2\sqrt{5}}}{4}$

$\cos^2 9 = \frac{\sqrt{10+2\sqrt{5}}+4}{8}$

$\Rightarrow \sin^2 81 = \frac{4+\sqrt{10+2\sqrt{5}}}{8}$

After squaring all the options available, we come to a conclusion that option

Q.15 Answer:

3

Solution:

$3\sin\theta = 5(1-\cos\theta) = 5 \times 2\sin^2\theta/2 \Rightarrow \tan\theta/2 = 3/5$

$5\sin\theta - 3\cos\theta = 5 \times \frac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}} - 3\frac{\left(1-\tan^2\frac{\theta}{2}\right)}{1+\tan^2\frac{\theta}{2}} = 5 \times \frac{2\times\frac{3}{5}}{1+\frac{9}{25}} - \frac{3\times\left(1-\frac{9}{25}\right)}{1+\frac{9}{25}} = 3$

Hence (b) is the correct answer.

Q.16 Answer:

$Q_1 = Q_3$

Solution:

We take Q_3 first, $Q_3 = \sin A(\cos B + \cos C) + \sin B(\cos C + \cos A)$
$+ \sin C(\cos A + \cos B)$
$= \sin A\cos B + \sin A\cos C + \sin B\cos C + \sin B\cos A$
$+ \sin C\cos A + \sin C\cos B$
$= \sin(A+B) + \sin(B+C) + \sin(C+A) = Q_1$
$\Rightarrow \quad Q_3 = Q_1$

Q.17 Answer:

1

Solution:

Given $A = (\cos 12^\circ - \cos 36^\circ)(\sin 96^\circ + \sin 24^\circ)$
$B = (\sin 60^\circ - \sin 12^\circ)(\cos 48^\circ - \cos 72^\circ)$
$\frac{A}{B} = \frac{[-2\sin 24^\circ \sin 12^\circ][2\sin 60^\circ \cos 36^\circ]}{[2\cos 36^\circ \sin 24^\circ][-2\sin 60^\circ \sin 12^\circ]}$
$\Rightarrow \frac{A}{B} = 1$

Q.18 Answer:

< 1

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Solution:

We have $\sin\alpha+\sin\beta+\sin\gamma-\sin(\alpha+\beta+\gamma)$
$=\sin\alpha+\sin\beta+\sin\gamma-\sin\alpha\cos\beta\cos\gamma$
$-\cos\alpha\sin\beta\cos\gamma-\cos\alpha\cos\beta\sin\gamma$
$+\sin\alpha\sin\beta\sin\gamma$
$=\sin\alpha(1-\cos\beta\cos\gamma)+\sin\beta(1-\cos\alpha\cos\gamma)$
$+\sin\gamma(1-\cos\alpha\cos\beta)+\sin\alpha\sin\beta\sin\gamma>0$
$\therefore\quad \sin\alpha+\sin\beta+\sin\gamma>\sin(\alpha+\beta+\gamma)$
Trick :Put $\alpha=30,\beta=30,\gamma=60$ **and check...**
$\Rightarrow \frac{\sin(\alpha+\beta+\gamma)}{\sin\alpha+\sin\beta+\sin\gamma}<1$

Q.19 Answer:

$-\frac{(4+\sqrt{7})}{3}$

Solution:

Given $\cos x+\sin x=\frac{1}{2}\Rightarrow 1+\sin 2x=\frac{1}{4}$
$\Rightarrow\quad \sin 2x=-\frac{3}{4}$, **so** x **is obtuse and**
$\frac{2\tan x}{1+\tan^2 x}=-\frac{3}{4}$
$\Rightarrow 3\tan^2 x+8\tan x+3=0$
$\therefore\quad \tan x=\frac{-8\pm\sqrt{64-36}}{6}=-\frac{-4\pm\sqrt{7}}{3}$
as $\tan x<0\quad \therefore\tan x=\frac{-4-\sqrt{7}}{3}$

Q.20 Answer:

1 and 3 only

Solution:

Let $A=30^\circ$
$\Rightarrow \sin A+2\sin 2A+\sin 3A$
$=\sin 30^\circ+2\sin 60^\circ+\sin 90^\circ$
$=\frac{1}{2}+\frac{2\sqrt{3}}{2}+1=\frac{2\sqrt{3}+3}{2}$
$\left(\because 2\cos^2 A=1+\cos 2A\right)$
Now, $4\sin 2A\cos^2\left(\frac{A}{2}\right)=2\sin 2A[1+\cos A]$
$=2\sin 60^\circ[1+\cos 30^\circ]=\frac{2\sqrt{3}+3}{2}$
Also, $\sin 2A=2\sin A\cos A\&\sin^2 A+\cos^2 A=1$
$2\sin 2A\left[\sin\frac{A}{2}+\cos\frac{A}{2}\right]^2$
$=2\sin 2A\left[\sin^2\frac{A}{2}+\cos^2\frac{A}{2}+2\sin\frac{A}{2}\cos\frac{A}{2}\right]$
$=2\sin 2A[1+\sin A]=2\sin 60^\circ[1+\sin 30^\circ]=\frac{3\sqrt{3}}{2}$
$\&8\sin A\cos A\cos^2\left(\frac{A}{2}\right)$
$=4\sin A\cos A[1+\cos A]$

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$= 4 \sin 30^\circ \cos 30^\circ \left[1 + \cos 30^\circ\right]$

$= \frac{2\sqrt{3}+3}{2}$

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